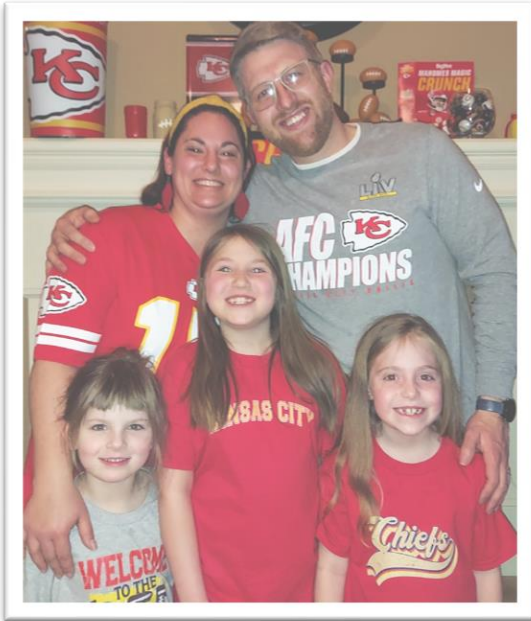


Precision Quoting for Avionics Installations



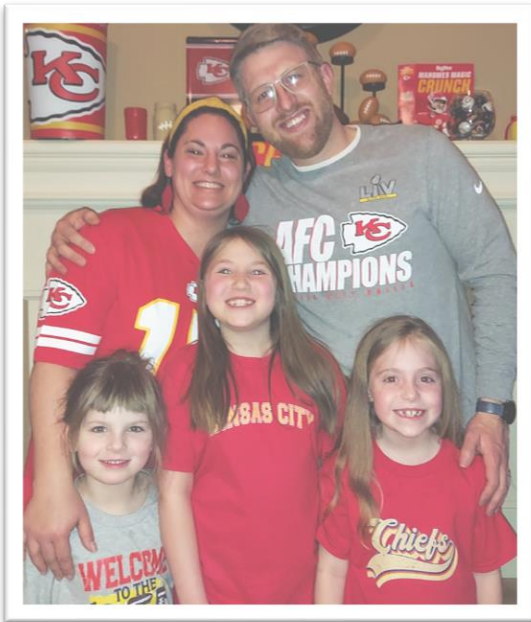
Introduction



Presenter: Levi Self

- Grew up in Southeast Kansas flying with my dad
- Aeronautical Engineering degree from LeTourneau University
- Former Garmin Systems Engineer and Project Manager.

Introduction



- Building RV-7, working on finish kit.
- CSEL/MEL/CFII/MEI, ~1800 hrs
- Founder and Owner of Midwest Avionics LLC, a Part 145 Repair station at KLXT.
- Co-Owner of Summit Flight Academy
- Repairman and A&P certificates
- Instructor for AEA training courses

Definition of a Quote

Definition of a Quote



A quote (or quotation) is an exact price for a job being offered. The price is fixed and should not be changed once it has been accepted by the customer unless:

- The customer changes the scope/type of work required.
- You discover an issue outside of the agreed scope of work.

Definition of a Quote

■ Quotes vs Estimates

- Although these terms are often used interchangeably, beware of the difference in language.
- An estimate is a best guess of how much a job might cost, while a quote is considered a fixed price
- Always be clear in your communication to the customer if you are giving an estimate or a quote.
- Don't be vague or misleading.

Why Give a Quote

- A quote sets clear expectation for your shop
 - A comprehensive quote helps you define the scope of the job including:
 - Planning
 - Scheduling
 - Ordering

Why Give a Quote

- A quote sets clear expectation for your customer.
 - The quote ensures the customer knows what is and isn't planned as part of the installation.
 - The customer needs to ensure he has sufficient funds
 - Customers don't like surprises.

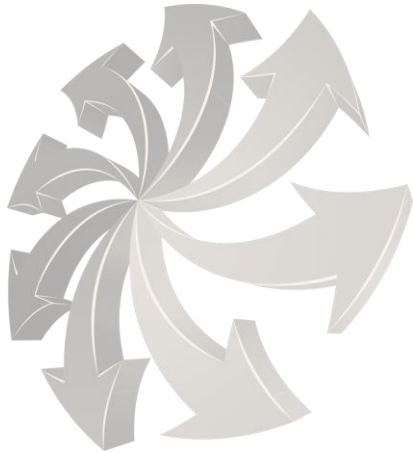
Tips to a Successful Quote

Tips to a Successful Quote



- **Determine if the job and customer fit your target market**
- If you suspect someone is not going to move forward with the quote, try to talk them out of the quote.
- Examples:
 - If their only concern is price.
 - If they need the job started before your availability.
 - If they want equipment you can't sell.
 - If you don't have capabilities to work on their aircraft.

Tips to a Successful Quote



- **Understand the scope of the job**
 - Spend enough time with the customer and airplane to gain a complete understanding of the job.
 - Spending time with the customer shows a level of care and interest.
 - Assess if the customer is asking for the “right” equipment for their aircraft.

Tips to a Successful Quote

- **Set your optimism aside!**
 - Don't think in the terms of, "if everything goes well..."
 - It is very unlikely that a whole job will go well.
 - Add labor to cover the inevitable challenges and oversights from the quote.
 - Going back to the customer each time you run into a small snag will cause you to lose credibility.

Tips to a Successful Quote

- **Track your shop metrics**
 - Track actual time to perform jobs to refine your future quotes.
 - Track acceptance rates of quote to watch trends
 - Track shop backlog over time

Tips to a Successful Quote



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- **Consider how badly you want the job**
 - If you are booked up, don't discount your time and materials.
 - If you are needing business, you may consider quoting a job at a lower rate or in fewer hours to help secure a bid.

Tips to a Successful Quote

- **Try to keep most of your work in your wheelhouse**
 - Try to guide the customer to equipment that you know well.
 - Become good at passing on work.
 - Grow your shop slowly and methodically. Don't take on more than one growth job at a time!
 - Growing too quickly and stretching your capabilities can quickly kill your profitability.

Tips to a Successful Quote

- How do you handle jobs you can't quote?

EXAMPLES:

- Troubleshooting an issue
- Small jobs with large amounts of integration into existing systems.
- Repairing an installation



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Tips to a Successful Quote

- How do you handle jobs you can't quote?
 - Set CLEAR expectations:
 - Ensure the Customer knows your pricing.
 - Set a price cap/hour limit where you will contact them if it is to be exceeded.
 - Keep the Customer updated during the process.
 - BE CAREFUL!



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Tips to a Successful Quote

BE LEARY OF A CUSTOMER THAT SAYS:

“I just want it done right, don’t bother with a quote, I trust you”

- It is much better to have a disagreement up front about a price being too high, than for you to have completed the work and them be upset about the price.

Tips to a Successful Quote

- **Be as specific as possible on the work to be performed**
 - The quote gives you the framework to start the job. All of the pertinent information to start the job should be contained within a quote.
 - If you have to rely on getting significant amounts of information from the customer after it is quoted, your quote is probably not detailed enough.
 - **EXAMPLE:** Do we need to source a new CDI to interface with the new radio?

Tips to a Successful Quote

- **Be as specific as possible on the work to be performed**
 - This helps to avoid customers from assuming what will be done.
 - When you have a detailed quote, you will not be on the hook to do work which you were not intending to do.
 - Example: Customer asks, "You were going to change that NAV antenna when you were putting the new radio in, right?"

Elements of a Quote

Elements of a Quote

- **CUSTOMER INFORMATION**

- Include all customer information including aircraft tail number
- This quote is specific to this customer and aircraft

- **Expiration Date**

- Place an expiration date to protect yourself

Elements of a Quote

■ PARTS

- Itemize out parts including connector kits, install kits, brackets, etc.
- List out basic functions with each part
 - EXAMPLE: GTN 650Xi, IFR GPS, VHF NAV, COM RADIO.
- As much as you are able, include small additional parts. E.g. circuit breakers, switches, jacks

Elements of a Quote

- **LABOR**

- It is better to lose a quote due to being too expensive, than to dramatically underbid a job and lose money on that job.

Elements of a Quote

- **LABOR**

- Disassembly of aircraft
- Removal of old equipment
- Removal of wiring
- Returning aircraft to Type Certificate configuration

Elements of a Quote

■ LABOR

- Creating wiring diagrams
- Building a wiring harness
- Installation of new equipment
 - Wiring in aircraft
 - Mounts and brackets
 - Switches and circuit breakers
 - Antennas and cables
 - Remote units
- Configuration and calibrations
- Paperwork

Elements of a Quote

- **Additional Labor considerations:**
 - Everything is not going to go according to plan.
 - Typically add 20%-25% to the labor hours estimate over what would normally be assessed, if I do not have good data for completing a job.
 - Add options for additional items not included in quote
 - Example: Rework circuit breaker bus with all new circuit breakers.
 - Add a new dimmer to dim radios

Elements of a Quote

A man with glasses and a beard is sitting at a desk in a workshop or office setting. He is looking at a laptop screen and has his hand resting on his chin, appearing to be in deep thought or listening intently. The background shows some industrial equipment and a window.

- **Additional Labor considerations:**
 - If you are the expert in your shop, don't quote the jobs as how long it would take you to complete the work.
 - Think about how long it would take an average technician to complete a task.

Elements of a Quote

- **Additional Labor considerations:**
 - Add extra labor to installations which require significant integration to existing equipment.
 - Assess if the existing installation is of acceptable quality. Any issues may be your responsibility.
 - Can you access the existing installation without significant disassembly.
 - Ensure you understand Interfaces to 3rd party equipment. Can you get the manuals and spec sheets for all of the equipment?

Elements of a Quote

- **Additional Labor considerations:**
 - In the Automotive Industry, there are labor guide books which give labor estimates for each task.
 - We don't have an internal labor guide as there are too many variables.
 - Each job is a custom installation
 - Avionics are not plug and play
 - Avionics installs in smaller GA aircraft are more like a custom hotrod build, versus a general auto repair.

Elements of a Quote

- **NOTES**

- Note what is not included in the quote
- If you are not planning to change a part of a system, be sure to note that in the quote.
 - Example: The COM antenna and coax cables are planned to be reused.

Elements of a Quote

- **FEES / EXTRA SERVICES**

- Be sure to add any additional fees or items on the quote, don't wait until the invoice.

Examples:

- 91.411/91.413 inspections
- Weight and Balance Fee if you are re-weighing the airplane.
- Administrative / Paperwork
- Shop Supply Fee



Elements of a Quote

■ **PLACEHOLDER FOR ADDITIONAL PARTS REQUIRED**

- Add a placeholder for additional parts not itemized on the quote.
- This usually works out to around 3-5% of the total cost of the job.
- Placeholder covers the cost for circuit breakers, switches, and other parts to keep the final invoice very close to the quote price.
- All items are itemized on the final invoice

Elements of a Quote

■ TIMELINES

- When scheduling work, we use a rule of thumb of 2x shop time for each billed hour.
- Be careful giving exact timelines unless your shop is specifically set up to support them.
- Any date you give the customer they will latch on to.
- Beware of performance guarantees and other terms.

Elements of a Quote

- **FINE PRINT**

- Any disclaimers or legal fine print should be included on the quote.
 - Acceptable forms of payment
 - Fees
 - Disclaimers
- If a deposit is required, note the deposit amount in the quote.
- Explain what is required to hold a position in your schedule

Elements of a Quote

- **CHANGE ORDERS/CHANGES**

- Any changes made to the scope of work should be added to quote
- Each change requires the customer to accept the updated quote
 - This ensures customers know what the total price will be.

Upselling

- The term upselling has a negative connotation, but done right, it will actually make the customer happier and make your shop more profitable.
 - EXAMPLE #1 If a customer asks for a new audio panel and a new COM, consider offering a whole radio stack as an option.
 - EXAMPLE #2 If a customer wants a new radio, consider also replacing the indicator as well to minimize rewire and ensure there isn't a parts issue.
 - The more wiring you can do outside the aircraft, the faster the job will be completed.
 - If wiring to an existing system, ensure that system is fully functional.
 - Selling additional equipment will be more profitable.

Upselling

- Ask new customers the date of the last 91.411/91.413 inspections.
 - Often they don't know and you can help them gain compliance while they are already in your shop.
- Offer to reweigh the aircraft
- Prioritize jobs that are profitable.
 - Lead customers toward jobs which are more simple and profitable.

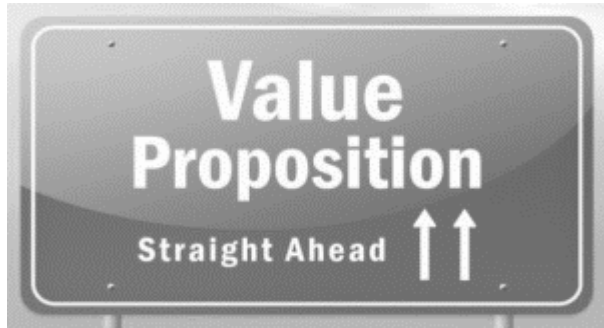
Closing the Deal

- Work to build trust with the customer through the entire process.
- Always be completely honest, customers can sniff out small lies.
- Go out of your way to be helpful.

Closing the Deal

- Follow-up after the quote is sent to ensure they received it.
- If possible, show the customer your work on another aircraft.
- Take the customer flying with the equipment they are looking to install.

Closing the Deal



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- Understand your value proposition and sell the customer on your business.
 - For Midwest Avionics, we run in the top 25% of quote prices, but customers still choose to do business with us.
 - “If you come here, when you leave everything works right”
 - “We have very little rework”
 - “We hit schedule dates”
 - “We do high quality work”

Closing the Deal

- If written acceptance is required, document how you would like to receive the approval.
- The expiration date give a limited amount of time to sign.
 - This can give a sense of urgency.
 - Follow up before quote expiration to see if they want to move forward.
 - If they decline, this is a great opportunity to understand why.

Closing the Deal

- Let people know if a price change is coming
 - Many people will push to accept the quote prior to a price change to lock in the labor rate.
- Be honest with customers about limitations.
 - Customers may be unhappy about an honest answer, but it will ultimately build trust with the customer.

Pay special attention!

CAUTION

- Be careful quoting jobs you don't completely understand.
- Don't repeat quotes before you have done a similar job that is complete.

Pay special attention!

CAUTION

- Beware of the “small asks”. They normally start with “while you are in there”
 - Adding lighting, or a USB charger or other item may seem very small, but can add up quickly.
- Creating a panel overlay or moving units can take significant time.

Pay special attention!

CAUTION

- **DON'T PRESUME YOU KNOW HOW YOUR CUSTOMER THINKS**
 - Be careful putting yourself in the customer's shoes. You may be in a different financial situation or you may have different desires from the customer. Don't presume to know what the customer values.
 - I might not do something to my aircraft, but maybe it makes sense to a customer.

Pay special attention!

CAUTION

- **BEWARE OF DIFFERENCES BETWEEN AIRFRAMES**

- Autopilot installations can vary dramatically from airframe to airframe.
- Sometimes we only consider the equipment to be installed and the airframe is an afterthought.
- If working on a retractable gear aircraft, be aware of risks.
 - You will need to swing gear to test magnetometers.
 - Always jack up a retract gear airplane when powering on a system if the gear system was touched during installation.

Pay special attention!

CAUTION

- **BEWARE OF DIFFERENT AIRFRAMES**

- Pressurized airplanes have extra considerations.
 - Even if there are wiring passthroughs, dealing with sealant can be very time consuming.
- Ensure STCs cover the aircraft and the configuration of the aircraft.
 - Some STCs are mutually exclusive.
- Understand interfaces to systems already in the aircraft.
- Beware of unsupported airframes and configurations.

QUESTIONS?